

Shahzad Ali

 **Work :** Genova, Italy

 **Email:** shahzadali039@gmail.com  **Email:** shahzad.ali6@unibo.it  **Phone:** (+39) 3482657723

 **Website:** <https://alishahzi.github.io/>  **WeChat:** shahzad-aliii

 **LinkedIn:** www.linkedin.com/in/shahzadali21  **Scopus AuthorID:** 57202066536

 **WoS ResearcherID:** ODJ-8978-2025  **ResearchGate:** Shahzad-Ali-56  **ORCID:** 0000-0002-0608-9515

Gender: Male **Date of birth:** 12/08/1992

ABOUT ME

I'm a passionate PhD scholar driving Graph Learning and AI (Machine/Deep Learning) for neurodegenerative disease modelling in healthcare. My work leverages Graph Neural Networks (GNNs) and multimodal data-driven methods to address clinically relevant challenges, supported by strong experience in large-scale datasets, algorithm development, and predictive modelling. I have presented my research in international journals and conferences, and I enjoy collaborating with peers and mentoring students.

EDUCATION AND TRAINING

- [01/11/2022 – Current]

PhD Data Science and Computation

Alma Mater Studiorum Università di Bologna <https://www.unibo.it/it>

City: Bologna | **Country:** Italy | **Field(s) of study:** Machine Learning Methods in Digital Health
- [15/07/2025 – 15/01/2026]

Visiting Researcher

Sussex AI Centre, University of Sussex, England <https://www.sussex.ac.uk/>

City: Brighton and Hove | **Country:** United Kingdom | **Field(s) of study:** Artificial Intelligence
- [01/2016 – 07/2018]

Master of Science in Computer Science

University of Engineering and Technology, Pakistan <https://uet.edu.pk/>

City: Lahore | **Country:** Pakistan | **Field(s) of study:** Computer Science | Machine Learning | **Final grade:** 3.75 / 4.00 | **Thesis:** Artificial Neural Network Model for Rainfall-Runoff Prediction for Jhelum River Basin Upstream of the Mangla Dam, Pakistan
- [08/2011 – 06/2015]

Bachelor of Science in Telecommunication and Networking

COMSATS University Islamabad, Sahiwal Campus <https://sahiwal.comsats.edu.pk/>

City: Sahiwal | **Country:** Pakistan | **Field(s) of study:** Computer Science | Telecommunication and Networking | **Thesis:** Visible Light-Based Communication System for High Data Rate Real-Time Applications

WORK EXPERIENCE

- IRCCS Ospedale Policlinico San Martino*

City: Genova | **Country:** Italy

Link <https://www.ospedalesanmartino.it/it/>
- [11/09/2023 – Current]

PhD Researcher

- Applied Machine Learning, Deep Learning, and Graph Networks for disease diagnosis using Neuroimaging data specifically focusing on multimodal data
 - Developed AI models to enhance diagnostic accuracy for neurological disorders.
 - Analyzed large-scale neuroimaging datasets to identify biomarkers and disease patterns.

- Collaborated with medical professionals and researchers for data-driven insights.
- Published research in top-tier journals and presented at international conferences.

Department of Information Sciences, University of Education <https://ue.edu.pk/>

City: Lahore | **Country:** Pakistan

[03/2019 – Current] **Lecturer - Computer Science**

Currently, teaching the following subjects at undergraduate level:

- Data Mining
- Artificial Intelligence
- Introduction to Machine Learning

Supervising final-year projects at undergraduate-level

Attending and speaking at conferences and seminars

Department of Computer Science & Engineering, University of Engineering and Technology, Lahore <https://uet.edu.pk/>

City: Lahore | **Country:** Pakistan

[10/2016 – 05/2018] **Graduate Research Assistant**

Taught and designed laboratory subject:

- Artificial Intelligence (Fall 2016, Fall 2017);

Assisted faculty members in creating and marking quizzes/assignments of undergraduate students and supervising final year and class projects.

PUBLICATIONS

[2026] [Exploring the spatial covariance of cerebral vascular density and amyloid burden in Alzheimer's disease](#)

Authors: Argenti L, Cirone A, Massa F, Sentieri E, Losa M, Lorenzini L, Garbarino S, Sofia L, Raffa S, Bauckneht M, Sambuceti G, Kreshpa W, Bozzo G, Tomassini G, De Cesari F, Pelagotti V, Pulze M, Gualco L, Hamedani M, Razzetta C, Brugnolo A, Girtler N, Caneva S, Mattioli P, Morbelli S, Ali S, Piana M, Serrati C, Uccelli A, Chincari A, Arnaldi D, Roccatagliata L, Bandini F, Orso B, Pardini M

Journal Name: Alzheimer's & Dementia: Diagnosis, Assessment & Disease Monitoring

[2026] [A robust multi-location evaluation of a machine learning framework for wind power forecasting](#)

Ali U, Sufyan M, Ali S, Ahmad M, Rehman Khan Su, Raza N, et al. (2026) A robust multi-location evaluation of a machine learning framework for wind power forecasting. PLoS One 21(4): e0344971. <https://doi.org/10.1371/journal.pone.0344971>

Authors: Usman Ali, Muhammad Sufyan, Shahzad Ali, Mudassar Ahmad, Sajawal ur Rehman Khan, Naeem Raza, Jabeen Sultana, Muhammad Asif Habib | **Journal Name:** PLOS ONE

[2026] [Neurotransmitter landscape and neurodegeneration patterns in Alzheimer's Disease](#)

Reference: Argenti, Lucia, Federico Massa, Mattia Losa, Lorenzo Lombardo, Luigi Lorenzini, Enrico Peira, Luca Sofia et al.

Journal Name: Neurobiology of Aging (Impact Factor: 3.5)

[2026] [A systematic study on the integration of MRI connectivity metrics for Alzheimer's diagnosis, staging, and cognitive decline prediction](#)

[2026] [Perception of Dental Faculty Regarding Development of Preclinical Endodontic Simulation Curriculum for Undergraduate Dental Students in Pakistan](#)

Naveed Manji, Salima, Imtiaz, Muhammad, Ehsan, Saroosh, Mehmood, Khizer, Ahmad, Shahzad, Zaheer, Naauman, Iqbal, Shazia, Ali, Shahzad, Perception of Dental Faculty Regarding Development of Preclinical Endodontic Simulation Curriculum for Undergraduate Dental Students in Pakistan, Scientifica, 2026, 1988107, 13 pages, 2026. <https://doi.org/10.1155/sci5/1988107>.

Journal Name: Scientifica (Impact Factor: 3.1)

[2026] [Hybrid Kernel SVM and Boosting Approaches for Accurate Autism Spectrum Disorder Screening](#)

Ali, Usman, Shaiza Shahid, Misbah, Shahzad Ali, Sheraz Aslam, and Kainat Mustafa. (2026). Hybrid Kernel SVM and Boosting Approaches for Accurate Autism Spectrum Disorder Screening. In: Iglesias, A., Shin, J., Bhatt, N., Joshi, A. (eds) Information Systems for Intelligent Systems. ISBM 2025. Lecture Notes in Networks and Systems, vol 1744. Springer, Cham. https://doi.org/10.1007/978-3-032-12996-3_17

Journal Name: In World Conference on Information Systems for Business Management, ISBM 2025

[2026] [Socioeconomic Indicator-Based Crime Prediction in Pakistan Using Machine Learning Techniques](#)

Ali, Usman, Tariq Hussain, Muhammad Mujeeb Akbar, and Shahzad Ali. "Socioeconomic Indicator-Based Crime Prediction in Pakistan Using Machine Learning Techniques." *Information Systems for Intelligent Systems: Proceedings of ISBM 2025, Volume 8* 8 (2026): 225. https://doi.org/10.1007/978-3-032-13003-7_22

Journal Name: In World Conference on Information Systems for Business Management, ISBM 2025

[2025] [Graph Neural Networks in Alzheimer's Disease Diagnosis: A Review of Unimodal and Multimodal Advances](#)

Ali, Shahzad, Michele Piana, Matteo Pardini, and Sara Garbarino. "Graph neural networks in Alzheimer's disease diagnosis: a review of unimodal and multimodal advances." *Frontiers in Neuroscience* 19 (2025): 1623141.

Journal Name: Frontiers in Neuroscience (Impact Factor: 3.2)

[2025] [Cerebrospinal Fluid Biomarkers Profiling in Cerebral Amyloid Angiopathy and Relationship With Disease Phenotypes](#)

Losa, Mattia, Matteo Cotta Ramusino, Isabella Cama, Lorenzo Gualco, Ilaria Gandoglia, Federico Massa, Andrea Donniaquio et al. "Cerebrospinal Fluid Biomarkers Profiling in Cerebral Amyloid Angiopathy and Relationship With Disease Phenotypes." *Journal of the American Heart Association* (2025): e044784.

Journal Name: Journal of the American Heart Association (Impact Factor: 6.106)

[2025] [Evaluation of Post-Endodontic Pain Reduction Using Intracanal Cryotherapy in Symptomatic Apical Periodontitis](#)

Authors: Anam Fayyaz Bashir, Ussamah Waheed Jatala, Muhammad Amber Fareed, Sheryar Sheryar, Saadia Ahmad Chattha, Saima Razaq Khan, Shahzad Ahmad, Shazia Iqbal, Muhammad Sohail Zafar, and Shahzad Ali | **Journal Name:** Australian Endodontic Journal (Impact Factor: 1.5)

- [2025] [An AI-Enabled Framework for Transparency and Interpretability in Cardiovascular Disease Risk Prediction](#)
- Kiran, Isha, Shahzad Ali, Musaed Alhussein, Sheraz Aslam, and Khursheed Aurangzeb. "An AI-Enabled Framework for Transparency and Interpretability in Cardiovascular Disease Risk Prediction." *Computers, Materials & Continua* 82, no. 3 (2025).
- Journal Name:** Computers, Materials & Continua (Impact Factor: 1.7)
- [2024] [Integration of Graph Theory and Brain Microstructural Analysis with Machine Learning for Advanced Diagnosis of Alzheimer's Disease and Prediction of Cognitive Decline](#)
- Shahzad Ali, Wendy Kreshpa, Michele Piana, Matteo Pardini, Sara Garbarino (2024), Integration of Graph Theory and Brain Microstructural Analysis with Machine Learning for Advanced Diagnosis of Alzheimer's Disease and Prediction of Cognitive Decline, IV Annual Meeting RIN IRCCS Network - Symposia, 21-22 November 2024.
- [2020] [Streamflow forecasting by modeling the rainfall-streamflow relationship using artificial neural networks](#)
- Ali, Shahzad, and Muhammad Shahbaz. "Streamflow forecasting by modeling the rainfall-streamflow relationship using artificial neural networks." *Modeling Earth Systems and Environment* 6, no. 3 (2020): 1645-1656.
- Journal Name:** Springer's Modeling Earth Systems and Environment (Impact Factor: 2.9)
- [2019] [Entropy-Based Feature Selection Classification Approach for Detecting Phishing Websites](#)
- Authors:** Shahzad Ali, Muhammad Shahbaz, and Kashif Jamil | **Journal Name:** 2019 13th International Conference on Open Source Systems and Technologies (ICOSST)
- [2019] [Classification of Alzheimer's Disease using Machine Learning Techniques](#)
- Authors:** Muhammad Shahbaz, Shahzad Ali, Aziz Guergachi, Aneeta Niazi, and Amina Umer | **Journal Name:** Proceedings of the 8th International Conference on Data Science, Technology and Applications: DATA
- [2019] [Prediction of Diabetes Disease using Data Mining Classification Techniques](#)
- Authors:** Shahzad Ali, Dawood Saddique, Muhammad Usman Aslam, Muhammad Usman, Umair Maqbool, Shoaib Ejaz | **Journal Name:** Proceeding of Al Yamamah University Engineering Forum (YUENG)

PROJECTS

- [02/2017 – 06/2018] **Artificial Neural Network Model for Rainfall-Runoff Prediction for Jhelum River Basin Upstream of the Mangla Dam, Pakistan**
- Master's Degree Thesis
- [08/2014 – 06/2015] **Visible Light-Based Communication System for High Data Rate Real-Time Applications**
- Bachelor's Degree Final Year Project
- Funded by ICT R&D, Federal Government of Pakistan

HONOURS AND AWARDS

- [02/2016] **Punjab Educational Endowment Fund (PEEF) Scholarship Awarding institution:** Government of Punjab, Pakistan
- Talent-based scholarship awarded during Master's degree

Link: <https://peef.org.pk/>

- [08/2014] **Final Year Project (Visible Light-Based Communication System for High Data Rate Real-Time Applications) Funded by ICT R&D Program**
Awarding institution: Funded by ICT R&D Program
- [09/2011] **ICT R&D Scholarship Awarding institution:** ICT R&D, Federal Government of Pakistan
Merit-based scholarship awarded during Bachelor's degree

CONFERENCES AND SEMINARS

- [22/05/2025 – 22/05/2025] **Update on the Pathophysiology and Neuroimaging of Amyloid-Related Imaging Abnormalities (ARIA)**
Genova, Italy
Attended the seminar "Update on the Pathophysiology and Neuroimaging of ARIA (Amyloid-Related Imaging Abnormalities)," focused on the neuropathology, immunopathological mechanisms, and MRI characteristics of ARIA-E and ARIA-H in Alzheimer's disease patients undergoing monoclonal antibody immunotherapy.
Link: <https://registration.symposiacongressi.com/Login.asp?iCurPage=1&IDEvento=286>
- [21/01/2024 – 26/01/2024] **2024 Geilo Winter School on the topic of Graphs and Applications** Geilo, Norway
[2024 Geilo Winter School](#) on the topic of [Graphs and Applications](#), which took place in Geilo (Norway) from January 21-26, 2024. The event consisted of multiple lectures, a computational competition, and a poster session, totalling 21 hours of instruction and work.
Link: <https://www.sintef.no/projectweb/geilowinterschool/2024-winter-school/>
- [25/10/2023 – 27/10/2023] **IEEE Communication Society (ComSoc) eHealth Technical Committee (TC) PhD School**
Genova, Italy
Attended the [IEEE Communication Society \(ComSoc\) eHealth Technical Committee \(TC\) PhD School](#) held in Genoa, Italy. The school consisted of 15 hours of Lectures with a Final Test.
Link: http://130.251.18.53/IEEE_ComSoC_PhD_School/ROOT/pages/index.html
- [10/07/2023 – 15/07/2023] **Eastern European Machine Learning (EEML) Summer School** Košice, Slovakia
Attended the [Eastern European Machine Learning \(EEML\) Summer School](#) held at the Technical University of Kosice, Slovakia, 10-15 July. The school consisted of 40 hours of Lectures and hands-on sessions.
Link: <https://www.eeml.eu/previous-editions/eeml2023>
- [10/03/2019 – 14/03/2019] **Data Science and Machine Learning for Physicists** Islamabad, Pakistan
Invited Speaker at **International School on Physics & Allied Disciplines** at the **National Center of Physics**, Islamabad, Pakistan

TECHNICAL SKILLS

Programming Languages

- Python | C++ [Expert]
- R | Java [Intermediate]

Machine/Deep Learning Frameworks and Tools

- PyTorch [Intermediate]
- Tensorflow | Keras | OpenCV [Intermediate]
- Sklearn [Expert]
- Jupyter Notebook | Google Colab [Expert]
- PowerBI | Tableau [Expert]
- Rapidminer Studio | Weka | KNIME [Expert]

Medical Imaging Tools

Hands-on with neuroimaging tools, specifically for neuroimaging and analysis including, FreeSurfer, MRtrix3, FSL, SPM, Clinica, Nilearn.

LANGUAGE SKILLS

Mother tongue(s): Urdu

Other language(s):

English

LISTENING C2 READING C1 WRITING C1

SPOKEN PRODUCTION C1 SPOKEN INTERACTION C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

RECOMMENDATIONS

will be furnished upon request.

Genova, 18/07/2026